

Tier-Prime Scaffold Entropy (TPS)

0. Purpose

We formalize an entropic framework where **geometry/meaning** comes from a learned **tier embedding** (semantics/cost), while **primes** serve only as a **canonical indexing scaffold** for tiers. The resulting descriptors support (i) within-window compressibility and (ii) cross-window novelty/stability.

1. Data model

Let Σ be a finite (or countable) symbol set. A data stream is $(X_t)_{t \in \mathbb{Z}}$ with $X_t \in \Sigma$. A **window** of length n ending at time t is

$$x_t := (X_{t-n+1}, \dots, X_t) \in \Sigma^n.$$

Let $c_s(x)$ denote the count of symbol s in window x .

2. Learned embedding and tier coordinate

2.1 Embedding

Let $f_\theta : \Sigma \rightarrow \mathbb{R}^d$ be an embedding map and define

$$z(s) := f_\theta(s) \in \mathbb{R}^d.$$

We require an **ordered scalar coordinate** $\tau : \Sigma \rightarrow \mathbb{R}$. Two standard constructions:

(A) Cost-supervised tier (if costs $E(s)$ are available):

- Learn f_θ and a head g to predict costs.

$$\tau(s) := \hat{E}(s) = g(z(s)).$$

(B) Self-supervised / unsupervised tier (if only sequences are available):

- Learn f_θ from context, then choose a 1D projection (e.g., PCA axis) $u \in \mathbb{R}^d$ and set

$$\tau(s) := u^\top z(s).$$

Assumption (Tier identifiability): τ is chosen so that nearby τ -values correspond to semantically/cost-similar symbols.

3. Tier quantization

Fix an integer $K \geq 2$ and thresholds

$$-\infty = b_0 < b_1 < \dots < b_{K-1} < b_K = +\infty.$$

Define the **tier assignment**

$$\kappa(s) = k \iff \tau(s) \in [b_{k-1}, b_k), \quad k \in \{1, \dots, K\}.$$

For a window x , define tier-counts

$$C_k(x) = \sum_{j=1}^n \mathbf{1}\{\kappa(x_j) = k\}, \quad n = \sum_{k=1}^K C_k(x),$$

and tier-frequencies

$$w_k(x) = \frac{C_k(x)}{n}.$$

Choice of thresholds. Common options:

- **Equal-mass bins** under a baseline distribution $q(s)$ (e.g., long-run frequencies) so each tier has comparable prior mass.
- **Equal-width bins** in τ -space.
- **Learned bins** that optimize downstream objectives (compression/anomaly labels).

4. Prime indexing scaffold

Let p_k be the k -th prime: $p_1 = 2, p_2 = 3, \dots$

4.1 Tier-coded integer (optional algebraic summary)

Define

$$N(x) = \prod_{k=1}^K p_k^{C_k(x)}.$$

This representation is **not** used for factorization; it is a commutative bookkeeping device. All computable statistics depend only on C_k (or w_k).

5. Within-window entropy: type/multiplicity entropy

Define the **tier type class size** (multinomial coefficient)

$$|T(C(x))| := \frac{n!}{\prod_{k=1}^K C_k(x)!}.$$

Define **type/multiplicity entropy**

$$H_T(x) = \frac{1}{n} \log |T(C(x))| = \frac{1}{n} \log \frac{n!}{\prod_{k=1}^K C_k(x)!}.$$

5.1 Asymptotics (Shannon emergence)

Let $w_k = C_k/n$. By Stirling, for large n ,

$$H_T(x) = -\sum_{k=1}^K w_k \log w_k + O\left(\frac{\log n}{n}\right).$$

Thus H_T approaches Shannon entropy of the tier distribution.

6. Tier spread (geometry-driven disorder)

To measure dispersion along the learned axis, assign each tier a representative coordinate $\tilde{\tau}_k$. Common choices:

- midpoint $\tilde{\tau}_k = (b_{k-1} + b_k)/2$ when finite,
- conditional mean $\tilde{\tau}_k = \mathbb{E}[\tau(S) \mid \kappa(S) = k]$ under a baseline distribution,
- empirical mean of $\tau(s)$ for symbols currently mapped to tier k .

Define the tier mean and variance in a window

$$\mu_\tau(x) := \sum_{k=1}^K w_k(x) \tilde{\tau}_k, \quad V_\tau(x) := \sum_{k=1}^K w_k(x) (\tilde{\tau}_k - \mu_\tau(x))^2.$$

Interpretation: V_τ measures **tier-mixing** across cost/semantic levels. Low V_τ indicates concentration in a narrow band.

7. Dynamics: novelty / stability via 1D transport

Define the induced discrete distribution on tier coordinates

$$\rho_x := \sum_{k=1}^K w_k(x) \delta_{\tilde{\tau}_k}.$$

Define the **spectral change** between adjacent windows (or any pair)

$$\Delta_\tau(t) := W_1(\rho_{x_t}, \rho_{x_{t-1}}),$$

where W_1 is the 1-Wasserstein distance on $\mathbb{R}$.

Computation (1D). In one dimension, W_1 can be computed from cumulative distributions; for discrete tiers this reduces to a simple cumulative-sum formula.

Interpretation: Δ_τ is a sensitive regime-shift statistic capturing **mass transport** along the tier axis.

8. Combined score (optional scalarization)

If a single scalar is desired, define for $\alpha \in [0, 1]$

$$\mathrm{TPS}^\alpha(x) := \alpha H_T(x) + (1-\alpha) \sqrt{V_\tau(x)}.$$

For dynamics, use $(H_T(x_t), \sqrt{V_\tau(x_t)}, \Delta_\tau(t))$ as a 3D descriptor.

9. Properties and invariances

1. **No symbol relabeling sensitivity (conditional):** If τ is learned/defined from semantics/cost and κ depends only on τ , then permuting symbol names does not change $(H_T, V_\tau, \Delta_\tau)$.
 2. **Prime scaffold irrelevance:** Primes do not affect $(H_T, V_\tau, \Delta_\tau)$; they only provide canonical labels $k \mapsto p_k$ for algebraic bookkeeping.
 3. **Shannon emergence:** H_T converges to the Shannon entropy of the tier-frequency vector w for large n .
 4. **Geometric refinement:** Two windows with identical w (hence identical H_T) can still differ in V_τ if tiers are defined differently (via learned τ), and can differ in Δ_τ across time.
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10. Practical validation protocol (minimal)

10.1 Synthetic separation

Construct windows with (approximately) equal tier Shannon entropy but different tier placement/mixing along $\tilde{\tau}$. Verify V_τ and Δ_τ separate cases where H_T does not.

10.2 Compression correlation

On real streams, compute sliding-window H_T and V_τ , and correlate with achieved compression ratio (e.g., zstd/gzip) under fixed windowing. Expect (H_T, V_τ) to outperform H_T alone for short windows.

10.3 Anomaly / regime shifts

Apply change-point detection to Δ_τ (and optionally $\sqrt{V_\tau}$). Inject controlled novelty events and evaluate detection delay and false positives.

11. Notes on design choices

- **Choice of K :** Larger K gives finer geometry but increases variance of estimates in short windows.
 - **Tier drift:** If τ is learned online, keep tiers stable via slow updates or anchoring to a baseline distribution.
 - **Embedding updates:** Separate timescales: fast window statistics; slow embedding/tier updates.
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12. Summary

TPS replaces assigned primes with a learned tier geometry τ (semantics/cost). The system is summarized per window by:

- H_T : microstate multiplicity of tier counts (type entropy),
- V_τ : dispersion across the learned tier axis,
- Δ_τ : temporal mass transport along tiers (novelty/stability). Primes remain only as a canonical indexing scaffold for tiers, optionally yielding a compact integer summary $N(x) = \prod p_k^{C_k}$.

References

1. Apostol, T. M. (1976). *Introduction to Analytic Number Theory*. Springer. (See Ch. 1: The Fundamental Theorem of Arithmetic.)
2. Hardy, G. H., & Wright, E. M. (2008). *An Introduction to the Theory of Numbers* (6th ed.). Oxford University Press.
3. Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379–423, 623–656.
4. Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.
5. Csiszár, I. (1998). The method of types. *IEEE Transactions on Information Theory*, 44(6), 2505–2523.
6. Csiszár, I., & Körner, J. (2011). *Information Theory: Coding Theorems for Discrete Memoryless Systems* (2nd ed.). Cambridge University Press.
7. Rissanen, J. (1978). Modeling by shortest data description. *Automatica*, 14(5), 465–471.
8. Polyanskiy, Y., Poor, H. V., & Verdú, S. (2010). Channel coding rate in the finite blocklength regime. *IEEE Transactions on Information Theory*, 56(5), 2307–2359.
9. Kostina, V., & Verdú, S. (2012). Fixed-length lossy compression in the finite blocklength regime. *IEEE Transactions on Information Theory*, 58(6), 3309–3338.
10. Villani, C. (2009). *Optimal Transport: Old and New*. Springer.
11. Peyré, G., & Cuturi, M. (2019). Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6), 355–607.
12. Santambrogio, F. (2015). *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*. Birkhäuser.
13. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv:1301.3781.
14. Mikolov, T., Sutskever, I., Chen, K., Corrado, G., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. In *Advances in Neural Information Processing Systems (NeurIPS 2013)*.

15. Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP 2014* (pp. 1532–1543).
16. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186).
17. van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748.
18. Gutmann, M. U., & Hyvärinen, A. (2010). Noise-contrastive estimation: A new estimation principle for unnormalized statistical models. In *Proceedings of AISTATS 2010* (PMLR 9, pp. 297–304).
19. Han, T. S. (2003). *Information-Spectrum Methods in Information Theory*. Springer.
20. Page, E. S. (1954). Continuous inspection schemes. *Biometrika*, 41(1-2), 100–114.
21. Killick, R., Fearnhead, P., & Eckley, I. A. (2012). Optimal detection of changepoints with a linear computational cost. *Journal of the American Statistical Association*, 107(500), 1590–1598.

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